

LiquidGo

Liquid Waste Management Marketplace

Product Architecture & Concept Document

Ghana | Version 1.0 | 2025

1. Executive Summary

LiquidGo is an Uber-style digital marketplace connecting customers who need liquid waste removal services with nearby independent operators and fleet companies in Ghana. The platform handles service discovery, intelligent driver routing, transparent pricing, and commission-based payments — all via a web application.

The core problem being solved: liquid waste disposal in Ghana is fragmented and difficult to find. Customers have no reliable way to discover, compare, or book service providers. Operators have no efficient channel to find jobs. LiquidGo bridges this gap with technology.

2. Product Overview

2.1 Core Value Proposition

- Customers can request liquid waste removal on demand and get matched to the nearest available driver.
- Drivers and fleet operators get a steady stream of verified jobs without manual marketing.
- The platform earns a commission on every completed transaction.
- Transparent pricing eliminates negotiation friction.

2.2 Service Scope

Phase 1 (MVP) will focus strictly on liquid waste removal, including:

- Septic tank emptying (residential and commercial)
- Soak pit/cesspit emptying
- Industrial liquid waste collection

Solid waste, recycling, and other waste streams are out of scope for Phase 1.

2.3 Geographic Focus

Initial launch: Ghana (starting with major urban centres — Accra, Kumasi, Tema). The architecture is designed to scale nationally.

3. User Types & Roles

Role	Description	Key Capabilities
Customer	Homes, businesses, landlords needing liquid waste removal	Request service, view pricing, pay, rate driver
Independent Driver	Solo operator with own tanker/vehicle	Accept jobs, set availability, view earnings
Fleet Manager	Company admin managing multiple drivers	Manage driver roster, view fleet analytics, receive bulk payouts
Fleet Driver	Driver registered under a fleet company	Accept jobs assigned by fleet or platform
Platform Admin	LiquidGo operations team	Approve drivers, manage pricing, resolve disputes, view analytics

4. Core Service Request Flow

The following describes the end-to-end lifecycle of a service request:

1. Customer logs in and submits a service request — entering their location, waste type, and estimated tank/pit size.
2. The system queries all online drivers within a configurable radius and ranks them by proximity.
3. The pricing engine calculates a quote (Base Fee + Distance Rate + Volume Tier).
4. The customer reviews the price and confirms the booking.
5. The nearest available driver receives a push notification and accepts or declines the job.
6. If declined, the request cascades to the next nearest driver.
7. The driver travels to the customer location and performs the service.
8. The driver marks the job as complete in the app.
9. Payment is captured via Paystack (MoMo or card). Commission is deducted automatically.
10. Both customer and driver rate each other. Funds are released to the driver.

5. Pricing Model

5.1 Formula

$$\text{Trip Cost} = \text{Base Fee} + (\text{Distance Rate} \times \text{km}) + \text{Volume Tier Fee}$$

5.2 Rate Components

Component	Description	Who Sets It
Base Fee	Fixed charge per job — covers driver's show-up cost	Driver (within platform min/max)
Distance Rate	Cost per km from driver's location to customer	Platform Admin
Volume Tier Fee	Flat fee based on estimated tank size (Small / Medium / Large)	Platform Admin
Platform Commission	Percentage deducted from total before driver payout (10–20%)	Platform Admin

5.3 Volume Tiers (Example)

Tier	Estimated Size	Additional Fee (GHS)
Small	Up to 2,000 litres	50
Medium	2,000 – 5,000 litres	100
Large	5,000+ litres	200

Note: All rates are configurable by the Platform Admin and can be adjusted per region.

6. System Architecture

6.1 High-Level Components

Module	Description
Auth & Onboarding	Separate registration for customers, drivers, and fleet companies. Document upload and admin approval for drivers.
Service Request Engine	Handles job creation, driver matching by proximity, job assignment, and status tracking.
Pricing Engine	Calculates real-time quotes based on distance, volume tier, and driver base rates.
Driver Management	Availability toggle, real-time location updates, job history, earnings dashboard.
Fleet Management	Sub-dashboard for companies to manage their registered drivers and view fleet-wide analytics.
Payment Module	Paystack integration for MoMo and card payments. Escrow, commission split, driver payout.

Module	Description
Ratings & Trust	Post-job rating system for both customers and drivers. Flags for low-rated accounts.
Admin Dashboard	Driver approvals, pricing configuration, transaction overview, dispute resolution, analytics.
Notification System	Real-time alerts for job assignments, status changes, and payment confirmations.

6.2 Driver Matching Algorithm

When a customer submits a request, the matching engine:

- Queries all drivers with status = ONLINE within a configurable radius (default: 15km)
- Calculates straight-line distance from each driver to customer using their last reported coordinates
- Filters out drivers who have an active job or are on cooldown
- Ranks remaining drivers by distance (closest first)
- Sends a push notification to the top-ranked driver with a 60-second accept window
- If declined or no response, cascades to the next driver in the ranked list
- If no driver accepts within the cascade, the request is marked as UNFULFILLED and the customer is notified

7. Tech Stack

Layer	Technology	Rationale
Frontend (Web)	React + Tailwind CSS	Fast development, component reuse across customer and driver portals
Backend API	DjangoRestFramework	large ecosystem, good WebSocket support
Database	PostgreSQL	Relational data suits transactional use cases; supports geospatial queries (PostGIS)
Real-time	Socket.io (WebSockets)	Live driver location updates and job notifications
Maps & Routing	Mapbox	Driver proximity calculation, distance estimates, map display
Payments	Paystack	Native Ghana MoMo support, card payments, split payout capability

Layer	Technology	Rationale
File Storage	Clouinary or AWS S3	Driver document uploads (license, permits)
Hosting	VPS contabo	Simple deployment, affordable for MVP stage
Authentication		Stateless auth, secure password hashing

8. Core Database Schema

8.1 Key Tables

Table	Key Fields
users	id, name, email, phone, role (customer driver fleet_admin admin), created_at
drivers	id, user_id, vehicle_reg, license_number, status (pending approved suspended), base_fee, is_online, last_lat, last_lng
fleet_companies	id, name, owner_user_id, registration_number, approved_at
service_requests	id, customer_id, driver_id, status, waste_type, volume_tier, pickup_lat, pickup_lng, quote_amount, commission_amount, created_at
payments	id, request_id, paystack_ref, amount, status, method (momo card), paid_at
ratings	id, request_id, rated_by, rated_user_id, score (1-5), comment, created_at

9. Payment Flow (Paystack)

- Customer initiates payment via Paystack Popup (MoMo or card) on job confirmation.
- Paystack charges the customer and holds funds.
- On job completion, the platform triggers a Paystack Transfer to the driver's registered mobile money account.
- Commission is retained by the platform before the transfer.
- All transactions are logged in the payments table with Paystack reference IDs for reconciliation.
- Drivers must register their MoMo number during onboarding to receive payouts.

10. Driver & Fleet Onboarding

10.1 Independent Driver

- Register with name, phone, email, password
- Upload: National ID, driving license, vehicle registration certificate, EPA waste handling permit
- Enter vehicle details (type, capacity in litres, registration number)
- Set base fee
- Admin reviews and approves — driver is notified by SMS/email
- Driver can go online and start receiving jobs

10.2 Fleet Company

- Company admin registers with business name, registration number, contact details
- Admin approves the company account
- Fleet admin can then add individual drivers under their company
- Each driver still undergoes the same document verification process
- Fleet admin dashboard shows all active drivers, ongoing jobs, and earnings

11. MVP Build Sequence

Recommended development order to get to a testable product quickly:

Phase	Features	Priority
Phase 1 — Foundation	Auth, registration flows, driver onboarding + admin approval	Must Have
Phase 2 — Core Loop	Service request, driver matching, job status tracking (manual updates)	Must Have
Phase 3 — Pricing & Payments	Pricing engine, Paystack integration, commission split	Must Have
Phase 4 — Trust & Ratings	Post-job ratings, driver/customer profiles, low-rating flags	Should Have
Phase 5 — Fleet Module	Fleet company registration, fleet admin dashboard	Should Have
Phase 6 — Admin Dashboard	Full admin panel: approvals, analytics, dispute resolution	Should Have
Phase 7 — Real-Time GPS	Live driver tracking on map, real-time location updates	Nice to Have (Post-MVP)

Note: Real-time GPS tracking is a Phase 7 feature. For MVP, manual status updates (En Route → Arrived → Job Complete) are sufficient and significantly reduce development complexity.

12. Key Risks & Mitigations

Risk	Impact	Mitigation
Low driver supply at launch	High	Pre-recruit drivers before launch; offer incentives for early sign-ups
Drivers gaming the system	Medium	Require document verification; rating system; admin monitoring
Payment failures (MoMo)	High	Paystack webhook handling; retry logic; customer support flow
Inaccurate location data	Medium	Allow manual address entry as fallback; validate coordinates on submission
Commission rate pushback from drivers	Medium	Start at 10% and communicate value clearly; transparent earnings dashboard

Appendix: Glossary

Term	Definition
MoMo	Mobile Money — digital wallet payment system used widely in Ghana (MTN, Vodafone, AirtelTigo)
Paystack	A payment gateway supporting card and mobile money payments across West Africa
PostGIS	A PostgreSQL extension enabling geospatial (location-based) database queries
Escrow	Funds held by the platform after customer payment, released to driver only after job completion
EPA Permit	Environmental Protection Agency permit required for liquid waste handlers in Ghana
Volume Tier	A pricing category based on estimated waste volume (Small / Medium / Large)

Term	Definition
Fleet Manager	A company administrator who registers and manages multiple drivers under one account